

Software Requirements Specification

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Subscription Management System

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1.0. Introduction

1.1. Purpose

The purpose of this document is to present a detailed description of the Subscription Management System. It explains the purpose and features of the system, the interfaces of the system, what the system will do, the constraints under which it must operate, and how the system will react to external stimuli. This document is intended for both stakeholders and developers of the system and is proposed as the requirements baseline for the problem statement.

1.2. Scope of Project

This software system is a centralized web application that helps organizations manage subscription-based products, recurring billing plans, customers, quotations, invoices, taxes, discounts, payments, and reports. It streamlines the complete subscription lifecycle from product configuration to payment collection and reporting, and is suitable for SaaS businesses and other recurring-revenue businesses.

More specifically, this system is designed to allow an Admin to configure products, recurring plans, taxes, and discounts; to allow Internal Users to create and manage subscriptions, quotations, invoices, and payments on behalf of customers; and to allow Portal/customer users to view their own subscriptions, invoices, and payment status. The system automates recurring invoice generation, tracks payments against invoices, applies configured taxes and discounts, and provides dashboards and reports summarizing active subscriptions, revenue, payments, and overdue invoices.

1.3. Glossary

Term	Definition
Admin	User role with full system control and configuration rights, including product, plan, tax, discount setup, and the sole role permitted to create Internal Users.
Internal User	User role with limited operational access used to carry out day-to-day subscription, invoicing, and payment tasks.
Portal/User	Customer or subscriber role with access limited to their own subscriptions, invoices, and payments.
Product	An item or service that can be sold on a recurring basis, with its own sales price, cost price, and optional variants.
Variant	An attribute-based pricing customization of a Product (for example, Brand: Odoo, extra price 560).
Recurring Plan	A configuration that defines the billing rules for a subscription product, including price, billing period, and plan options such as auto-close, closable, pausable, and renewable.
Subscription	A record linking a Customer to a Recurring Plan and one or more Products, tracked through the states Draft, Quotation, Confirmed, Active, and Closed.
Quotation Template	A predefined template used during subscription creation to speed up setup, containing validity days, a recurring plan, and product lines.
Invoice	A billing document automatically generated from a subscription, tracked through the states Draft, Confirmed, and Paid.
Payment	A record of the settlement of an invoice, including payment method, amount, and date.
Discount	A fixed or percentage reduction applied to products or subscriptions, created only by an Admin, subject to minimum purchase, minimum quantity, validity dates, and usage limits.
Tax	A configurable percentage or type applied automatically to invoice lines.
Customer	An individual or organization that subscribes to one or more recurring plans and is the subject of quotations, invoices, and payments.
Dashboard	A summary view presenting analytics such as active subscriptions, revenue, payments, and overdue invoices.
Software Requirements Specification	A document that completely describes all of the functions of a proposed system and the constraints under which it must operate. For example, this document.
Stakeholder	Any person with an interest in the project who is not a developer.

1.4. References

IEEE. IEEE Std 830-1998 IEEE Recommended Practice for Software Requirements Specifications. IEEE Computer Society, 1998.

Subscription Management System — Problem Statement (hackathon brief), supplied by the organizing team.

Mockup: <https://app.excalidraw.com/l/65VNwvy7c4X/8hI7WFOAKhj>

1.5. Overview of Document

The next chapter, the Overall Description section, gives an overview of the functionality of the product. It describes the informal requirements and establishes a context for the technical requirements specification in the following chapter.

The third chapter, the Requirements Specification section, is written primarily for developers and describes in technical terms the details of the functionality of the product.

Both sections describe the same software product in its entirety but are intended for different audiences and thus use different language.

2.0. Overall Description

2.1 System Environment

The Subscription Management System has three active actors: Admin, Internal User, and Portal/User (customer). All three actors access the system through a web browser over the Internet.

The Admin configures Products, Recurring Plans, Taxes, Discounts, and Internal User accounts, and has full visibility into every module. The Internal User operates the day-to-day subscription lifecycle: creating subscriptions, generating quotations and invoices, and recording payments, but cannot configure system-wide settings such as discounts, which are Admin-only. The Portal/User authenticates to view their own subscriptions, invoices, and payment history only.

The system automatically generates invoices from active subscriptions on their billing schedule, applies configured taxes and discounts, and updates the Reporting module in real time as subscriptions, invoices, and payments change state.

2.2 Functional Requirements Specification

This section outlines the use cases for each of the three actors separately. The Portal/User has a small, read-oriented set of use cases; the Internal User carries out the operational subscription workflow; and the Admin has the broadest set of use cases, covering system configuration in addition to the operational workflow.

2.2.1 Admin Use Cases

The Admin has the following sets of use cases, covering configuration of the catalog and billing rules, and user administration.

Configuration use cases

Use case: Manage Products

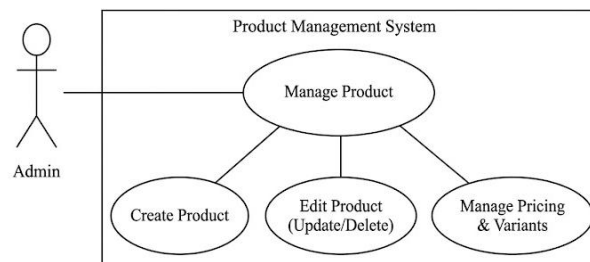
Before this use case can be initiated, the Admin has already logged in and accessed the Products Management screen.

Brief Description

The Admin creates, updates, deletes, or views a product, including its recurring pricing and any variants.

Initial Step-By-Step Description

1. The Admin selects Product Management from the navigation menu.
2. The system presents the list of existing products.
3. The Admin chooses to create a new product or selects an existing product to update or delete.
4. The Admin enters or edits the Product Name, Product Type, Sales Price, and Cost Price, and optionally adds variants (Attribute, Value, Extra Price).
5. The Admin submits the form.
6. The system validates and saves the product, and returns the Admin to the product list.



Expanded Manage Product Use Case Diagram - Simplified SRS format.

Use case: Manage Recurring Plans

Before this use case can be initiated, the Admin has already logged in and accessed the Recurring Plan Management screen.

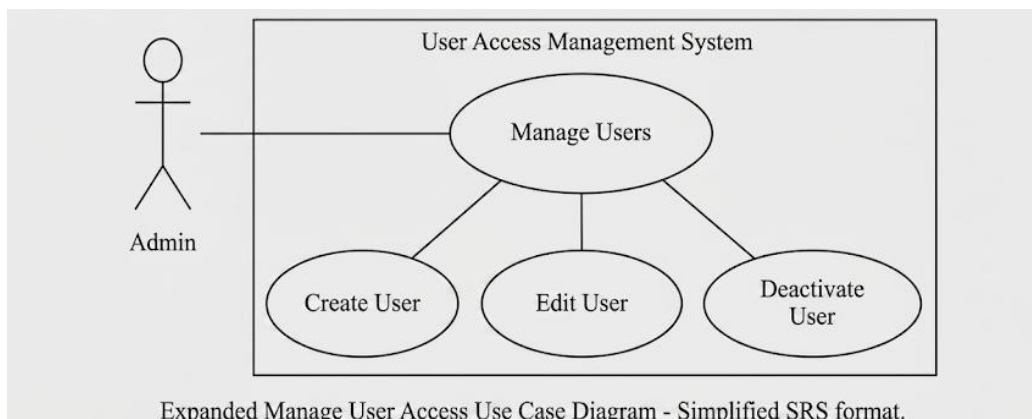
Brief Description

The Admin defines or edits the billing rules for a subscription product.

Initial Step-By-Step Description

7. The Admin selects Recurring Plan Management from the navigation menu.
8. The system presents the list of existing plans.
9. The Admin chooses to create a new plan or selects an existing plan to update.
10. The Admin enters the Plan Name, Price, Billing Period (Daily / Weekly / Monthly / Yearly), Minimum Quantity, Start Date, End Date, and plan options (Auto-close, Closable, Pausable, Renewable).
11. The Admin submits the form.
12. The system validates and saves the plan.

Xref: Section 3.2.4, Manage Recurring Plans



Use case: Manage Discounts

Before this use case can be initiated, the Admin has already logged in and accessed the Discount Management screen.

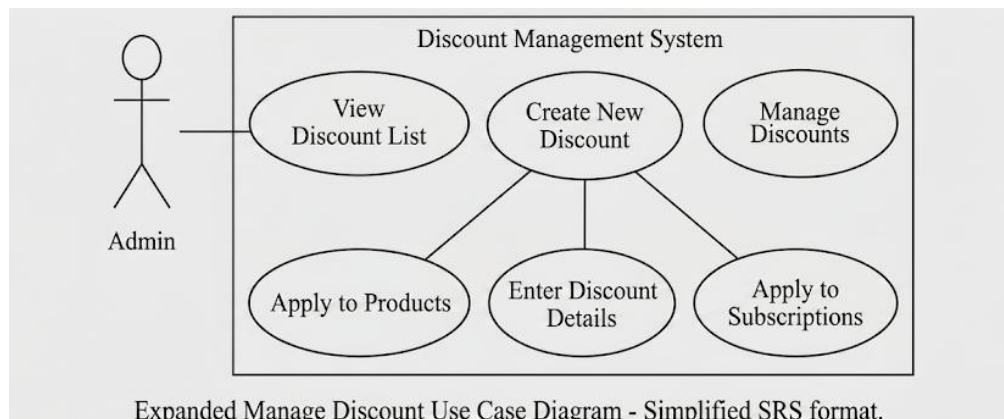
Brief Description

The Admin creates a discount rule to be applied to products or subscriptions. This use case is restricted to the Admin role.

Initial Step-By-Step Description

13. The Admin selects Discount Management from the navigation menu.
14. The system presents the list of existing discounts.
15. The Admin chooses to create a new discount.
16. The Admin enters the Discount Name, Type (Fixed or Percentage), Minimum Purchase, Minimum Quantity, Start Date, End Date, Limit Usage, and selects whether the discount applies to Products or Subscriptions.
17. The Admin submits the form.
18. The system validates and saves the discount.

Xref: Section 3.2.10, Apply Discount



Use case: Manage Taxes

Before this use case can be initiated, the Admin has already logged in and accessed the Tax Management screen.

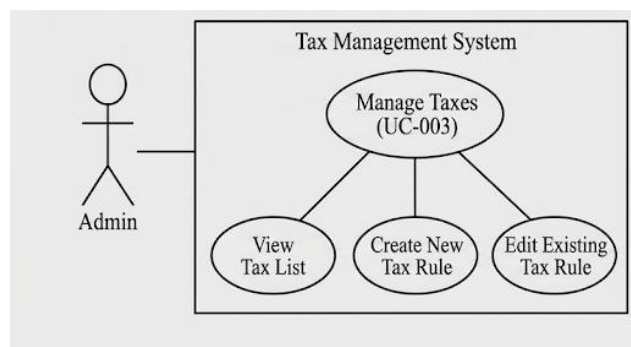
Brief Description

The Admin configures a tax percentage and type to be automatically applied during invoice generation.

Initial Step-By-Step Description

19. The Admin selects Tax Management from the navigation menu.
20. The system presents the list of existing tax rules.
21. The Admin chooses to create a new tax rule or edit an existing one.
22. The Admin enters the tax percentage and type.
23. The Admin submits the form.
24. The system validates and saves the tax rule.

Xref: Section 3.2.11, Apply Tax



User administration use case

Use case: Create Internal User

Before this use case can be initiated, the Admin has already logged in.

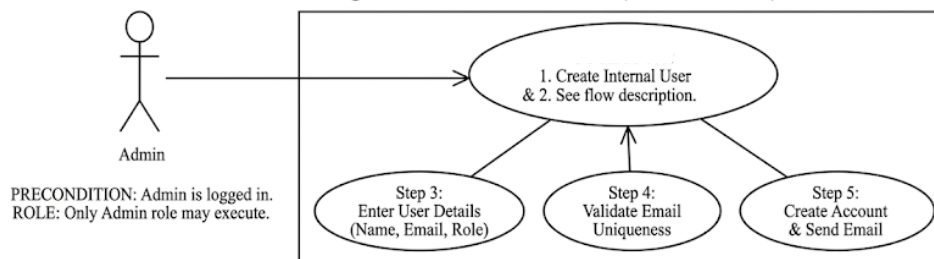
Brief Description

The Admin creates a new Internal User account. Only the Admin role may perform this use case.

Initial Step-By-Step Description

25. The Admin selects Users/Contacts from the navigation menu.
26. The Admin chooses to add a new Internal User.
27. The Admin enters the user's name, email, and role.
28. The system validates that the email is unique.
29. The system creates the Internal User account and sends account details to the registered email.

Xref: Section 3.2.1, Signup; Section 3.2.13, Manage Users



2.2.2 Internal User Use Cases

The Internal User operates the day-to-day subscription lifecycle on behalf of customers.

Use case: Create Subscription

Before this use case can be initiated, the Internal User has already logged in and accessed the Subscriptions screen.

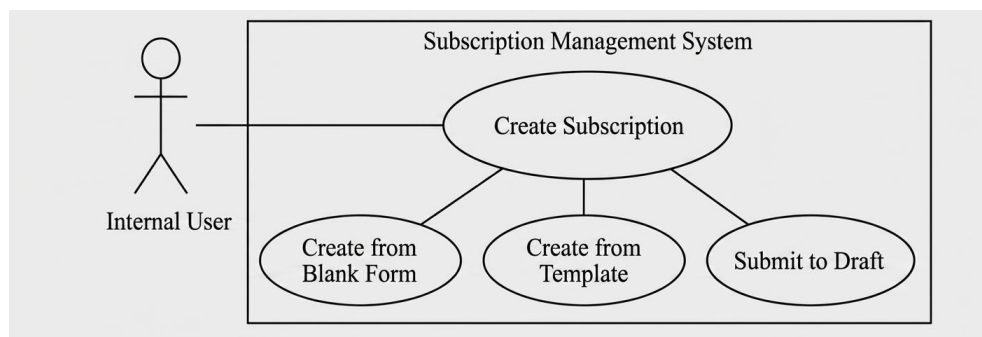
Brief Description

The Internal User creates a new subscription for a customer, either from scratch or from a quotation template.

Initial Step-By-Step Description

30. The Internal User selects Subscriptions from the navigation menu and chooses to create a new subscription.
31. The system presents a blank subscription form, or a form pre-filled from a chosen Quotation Template.
32. The Internal User selects the Customer, the Recurring Plan, the Start Date, and adds Order Lines (Product, Quantity, Unit Price, Taxes).
33. The Internal User submits the form.
34. The system creates the subscription in Draft status and computes line amounts, taxes, and totals.

Xref: Section 3.2.6, Create Subscription



Use case: Generate Invoice

Before this use case can be initiated, the Internal User has already accessed an active subscription.

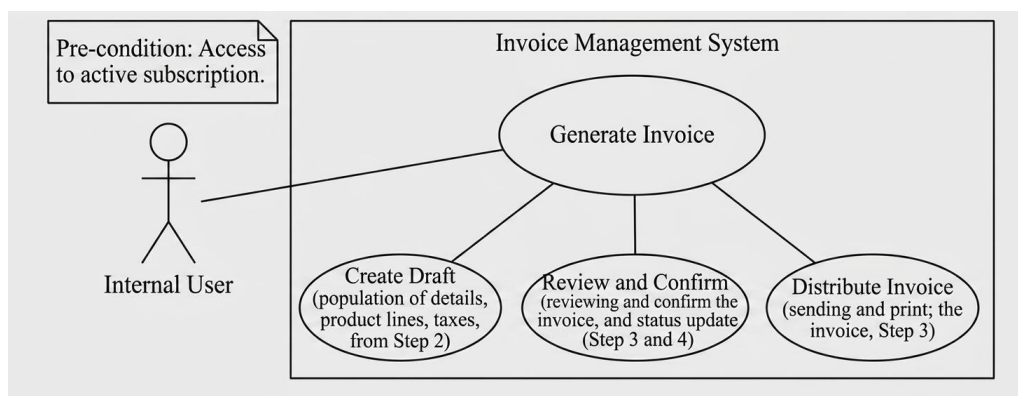
Brief Description

The system, on behalf of the Internal User, generates an invoice from an active subscription; the Internal User confirms, sends, or prints it.

Initial Step-By-Step Description

35. The Internal User selects an active Subscription and chooses to generate an invoice, or the system generates one automatically on the billing schedule.
36. The system creates a Draft invoice populated with customer details, product lines, taxes, and totals from the subscription.
37. The Internal User reviews the invoice and chooses to Confirm, Cancel, Send, or Print it.
38. The system updates the invoice status accordingly (Draft to Confirmed).

Xref: Section 3.2.8, Generate Invoice



Use case: Record Payment

Before this use case can be initiated, the Internal User has already accessed a confirmed invoice.

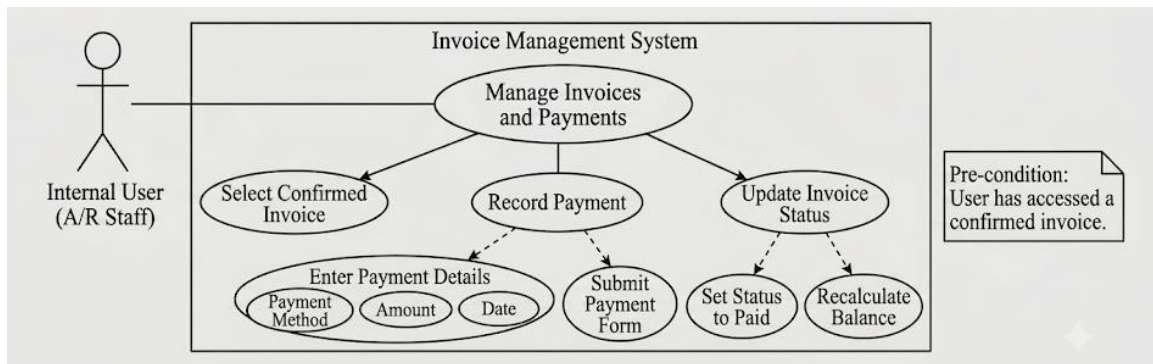
Brief Description

The Internal User records a payment against a confirmed invoice.

Initial Step-By-Step Description

39. The Internal User selects a Confirmed invoice and chooses to record a payment.
40. The system presents a payment form.
41. The Internal User enters the Payment Method, Amount, and Payment Date, and submits the form.
42. The system records the payment, updates the invoice status to Paid when fully settled, and updates outstanding balances.

Xref: Section 3.2.9, Record Payment



2.2.3 Portal/User (Customer) Use Cases

Use case: View Subscription and Invoices

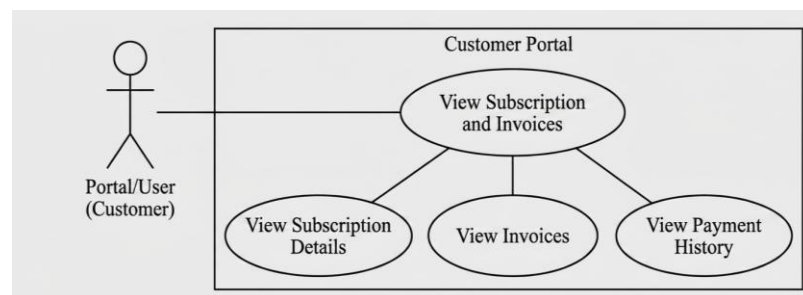
Before this use case can be initiated, the Portal/User has already been provisioned as a customer and has valid login credentials.

Brief Description

The Portal/User views their own subscription details, invoices, and payment history.

Initial Step-By-Step Description

43. The Portal/User logs in to the customer portal.
44. The system presents the customer's dashboard showing active subscriptions.
45. The Portal/User selects a subscription to view its details, order lines, and status.
46. The Portal/User navigates to Invoices to view invoice status (Draft, Confirmed, Paid) and download invoice documents.



2.3 User Characteristics

The Admin is expected to be familiar with business configuration concepts such as pricing, billing periods, taxes, and discounts, and to be comfortable with standard web-application forms, grids, and navigation menus.

The Internal User is expected to be trained on the day-to-day subscription workflow (creating subscriptions, generating invoices, recording payments) but does not require configuration-level knowledge.

The Portal/User is expected to be Internet literate and able to use a standard web browser to log in and view read-only account information; no specialized training is required.

2.4 Non-Functional Requirements

The system will be hosted on a server with reliable Internet connectivity, built as a web application with a browser-based front end and a REST-capable back end. Response time for standard operations must remain under 2 seconds, and the system must scale to support thousands of subscriptions.

Access to system functionality is governed by role-based permissions: only the Admin may create Internal Users or configure discounts; Internal Users have operational access to subscriptions, invoices, and payments; Portal/Users have read-only access limited to their own records. All authentication is performed over a secure connection, and passwords are validated for strength at signup and reset.

3.0. Requirements Specification

3.1 External Interface Requirements

The Subscription Management System exposes a web-based user interface for all three actor roles (Admin, Internal User, Portal/User), reached through a standard browser.

The system sends transactional email for account signup confirmation, password reset links, and invoice delivery (the Send action in Invoice Management). No other external system integration is required for the scope of this specification; payment gateway and accounting system integrations are out of scope for this version and may be added as external interfaces in a future revision.

3.2 Functional Requirements

The Logical Structure of the Data is contained in Section 3.3.1.

3.2.1 Login

Use Case Name	Login
XRef	Section 2.2, Authentication Module
Trigger	The user navigates to the login page and enters credentials.
Precondition	The user already holds a registered account with a unique email address.
Basic Path	1. The user enters email and password and submits the login form. 2. The system validates the credentials against the stored account. 3. The system creates a session and redirects the user to the role-appropriate

	dashboard (Admin, Internal User, or Portal).
Alternative Paths	None.
Postcondition	The user is authenticated and has an active session.
Exception Paths	If the email/password combination is invalid, the system displays an error and the user may retry or choose Reset Password.
Other	None.

3.2.2 Signup

Use Case Name	Signup
XRef	Section 2.2, Authentication Module
Trigger	A new user chooses to register for an account.
Precondition	The user is on the Signup page.
Basic Path	1. The user enters name, email, and password. 2. The system validates that the email is unique and that the password satisfies the strength rules. 3. The system creates the account and sends a confirmation email.
Alternative Paths	None.
Postcondition	A new user account exists in the system.
Exception Paths	If the email is already registered, or the password fails a strength rule (length > 8,

	uppercase, lowercase, special character), the system displays the specific validation error and the user may correct and resubmit.
Other	See Section 5.4 Validation Rules in the source problem statement for the full password policy.

3.2.3 Reset Password

Use Case Name	Reset Password
XRef	Section 2.2, Authentication Module
Trigger	The user chooses Forgot Password on the login page.
Precondition	The user has a registered account.
Basic Path	1. The user enters the registered email address. 2. The system verifies the email exists and sends a reset link to it. 3. The user follows the link and enters a new password meeting the strength rules. 4. The system updates the stored password.
Alternative Paths	None.
Postcondition	The account password is updated.
Exception Paths	If the email is not registered, the system displays a generic confirmation message without revealing account existence. The user may abandon the operation at any time.
Other	None.

3.2.4 Manage Products

Use Case Name	Manage Products
XRef	Section 2.2.1, Manage Products
Trigger	The Admin selects Product Management.
Precondition	The Admin has logged in.
Basic Path	1. The system presents the list of products. 2. The Admin chooses to create, update, delete, or view a product. 3. For create/update, the Admin enters Product Name, Product Type, Sales Price, Cost Price, and optional variants (Attribute, Value, Extra Price). 4. The system validates and saves the product.
Alternative Paths	If the Admin chooses Delete, the system asks for confirmation before removing the product.
Postcondition	The product catalog reflects the Admin's changes.
Exception Paths	The Admin may abandon the operation at any time. Deleting a product referenced by an active subscription is blocked with an explanatory message.
Other	Only the Admin role may access this use case.

3.2.5 Manage Recurring Plans

Use Case Name	Manage Recurring Plans
XRef	Section 2.2.1, Manage Recurring Plans
Trigger	The Admin selects Recurring Plan Management.
Precondition	The Admin has logged in.
Basic Path	1. The system presents the list of recurring plans. 2. The Admin chooses to create or update a plan. 3. The Admin enters Plan Name, Price, Billing Period, Minimum Quantity, Start Date, End Date, and plan options (Auto-close, Closable, Pausable, Renewable). 4. The system validates and saves the plan.
Alternative Paths	None.
Postcondition	The plan catalog reflects the Admin's changes.
Exception Paths	The Admin may abandon the operation at any time.
Other	Only the Admin role may access this use case.

3.2.6 Create Subscription

Use Case Name	Create Subscription
XRef	Section 2.2.2, Create Subscription
Trigger	The Internal User selects Subscriptions and chooses to create a new subscription.

Precondition	The Internal User has logged in; at least one Product and Recurring Plan exist.
Basic Path	1. The system presents a blank subscription form, or a form pre-filled from a chosen Quotation Template. 2. The Internal User selects Customer, Plan, Start Date, and Payment Terms, and adds Order Lines (Product, Quantity, Unit Price, Taxes, Amount). 3. The Internal User submits the form. 4. The system creates the subscription in Draft status and computes totals.
Alternative Paths	If created from a Quotation Template, product lines and the recurring plan are pre-populated from the template and may be adjusted before submission.
Postcondition	A new subscription exists in Draft status.
Exception Paths	The Internal User may abandon the operation at any time. Missing required fields (Customer, Plan) block submission with a validation message.
Other	Subscription status flow: Draft to Quotation to Confirmed to Active to Closed.

3.2.7 Generate Quotation from Template

Use Case Name	Generate Quotation from Template
XRef	Section 2.2.2, Create Subscription
Trigger	The Internal User selects an existing Quotation Template while creating a subscription.
Precondition	At least one Quotation Template has been configured.

Basic Path	1. The Internal User selects a Quotation Template from the list. 2. The system pre-fills the subscription form with the template's Recurring Plan, Product Lines, and Validity Days. 3. The Internal User reviews and adjusts the pre-filled data as needed and submits the form. 4. The system creates the subscription in Quotation status.
Alternative Paths	None.
Postcondition	A subscription exists in Quotation status, ready for confirmation.
Exception Paths	The Internal User may abandon the operation at any time.
Other	Template fields: Template Name, Validity Days, Recurring Plan, Product Lines.

3.2.8 Generate Invoice

Use Case Name	Generate Invoice
XRef	Section 2.2.2, Generate Invoice
Trigger	A subscription reaches its billing date, or the Internal User manually requests an invoice for an active subscription.
Precondition	The subscription is in Active status.
Basic Path	1. The system creates a Draft invoice populated with customer details, product lines, taxes, and totals from the subscription. 2. The Internal User reviews the invoice.

	3. The Internal User chooses Confirm, Cancel, Send, or Print. 4. The system updates the invoice status from Draft to Confirmed when confirmed.
Alternative Paths	If Send is chosen, the system emails the invoice to the customer's registered address. If Cancel is chosen, the invoice is voided and does not affect the subscription's billing schedule.
Postcondition	An invoice exists in Confirmed status (or Cancelled, per the alternative path).
Exception Paths	The Internal User may abandon the operation at any time before confirming.
Other	Invoice status flow: Draft to Confirmed to Paid.

3.2.9 Record Payment

Use Case Name	Record Payment
XRef	Section 2.2.2, Record Payment
Trigger	The Internal User selects a Confirmed invoice and chooses to record a payment.
Precondition	The invoice is in Confirmed status with an outstanding balance.
Basic Path	1. The system presents a payment form. 2. The Internal User enters Payment Method, Amount, and Payment Date. 3. The Internal User submits the form. 4. The system records the payment and updates the invoice status to Paid once the full amount is settled.

Alternative Paths	If the amount entered is less than the outstanding balance, the system records a partial payment and keeps the invoice in Confirmed status with a reduced outstanding balance.
Postcondition	The payment is recorded and the invoice's paid/outstanding amounts are updated.
Exception Paths	The Internal User may abandon the operation at any time.
Other	Payments are tracked against paid and outstanding invoices for reporting.

3.2.10 Apply Discount

Use Case Name	Apply Discount
XRef	Section 2.2.1, Manage Discounts
Trigger	An Admin-configured discount rule matches a product or subscription during quotation or invoice generation.
Precondition	At least one active discount rule exists, created by an Admin.
Basic Path	1. The system checks configured discount rules against the order lines being processed. 2. If a discount's Minimum Purchase, Minimum Quantity, and date-validity conditions are met, the system applies the discount (Fixed or Percentage) to the applicable Products or Subscriptions. 3. The system reduces the discount's remaining Limit Usage. 4. The system recalculates line and order totals.

Alternative Paths	If multiple discounts match, the system applies them per the configured priority (to be defined during design); if none match, no discount is applied.
Postcondition	Applicable order lines reflect the discounted amount.
Exception Paths	If a discount's Limit Usage has been exhausted or its validity window has passed, it is not applied.
Other	Discounts can be created only by an Admin.

3.2.11 Apply Tax

Use Case Name	Apply Tax
XRef	Section 2.2.1, Manage Taxes
Trigger	An invoice or subscription order line is generated or updated.
Precondition	At least one tax rule has been configured by an Admin.
Basic Path	1. The system determines the applicable tax rule(s) for each order/invoice line. 2. The system auto-calculates the tax amount using the configured percentage and type. 3. The system adds the tax amount to the line and updates the invoice total.
Alternative Paths	None.
Postcondition	Invoice lines and totals reflect the applicable tax.
Exception Paths	None.

Other	Tax percentages and types are configurable and auto-calculated during invoice generation.
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3.2.12 View Reports

Use Case Name	View Reports
XRef	Section 2.2, Reporting Module
Trigger	A user with reporting access navigates to the Reports/Dashboard screen.
Precondition	The user has logged in with a role permitted to view reports (Admin, Internal User; Portal/User sees only their own summary).
Basic Path	1. The system aggregates current data for active subscriptions, revenue, payments, and overdue invoices. 2. The system presents the dashboard/report view with summary figures. 3. The user applies filters (date range, status, customer) to narrow the summary view.
Alternative Paths	A Portal/User accessing this use case sees only figures scoped to their own account.
Postcondition	The requested report/summary is displayed.
Exception Paths	The user may abandon the operation at any time.
Other	Supports filtering and summary views as described in the Reporting Module.

3.2.13 Manage Internal Users

Use Case Name	Manage Internal Users
XRef	Section 2.2.1, Create Internal User
Trigger	The Admin selects Users/Contacts and chooses to add, update, or deactivate an Internal User.
Precondition	The Admin has logged in.
Basic Path	1. The system presents the list of Internal Users. 2. The Admin chooses to add a new Internal User, or selects an existing one to update or deactivate. 3. For a new user, the Admin enters name, email, and role, and the system validates the email is unique. 4. The system saves the change and, for a new user, emails account details to the registered address.
Alternative Paths	None.
Postcondition	The Internal User list reflects the Admin's changes.
Exception Paths	The Admin may abandon the operation at any time.
Other	Only the Admin can create Internal Users.

3.3 Detailed Non-Functional Requirements

3.3.1 Logical Structure of the Data

The Logical Structure of the data to be stored by the Subscription Management System is as follows:

User Data Entity

Data Item	Type	Description	Comment
Name	Text	Full name of the user	
Email	Text	Login identifier	Must be unique
Password Hash	Text	Securely stored password	Never stored or displayed in plain text
Role	Category	Admin / Internal User / Portal-User	

Product Data Entity

Data Item	Type	Description	Comment
Product Name	Text	Name of the product	
Product Type	Category	Classification of the product	
Sales Price	Currency	Price charged to the customer	
Cost Price	Currency	Internal cost of the product	
Variant	Pointer	Attribute-based pricing customization	May be several; see

Data Item	Type	Description	Comment
			Variant entity

Variant Data Entity

Data Item	Type	Description	Comment
Attribute	Text	Name of the varying attribute (e.g., Brand)	
Value	Text	Value of the attribute (e.g., Odoo)	
Extra Price	Currency	Additional price for this variant	

Recurring Plan Data Entity

Data Item	Type	Description	Comment
Plan Name	Text	Name of the plan	
Price	Currency	Base recurring price	
Billing Period	Category	Daily / Weekly / Monthly / Yearly	
Minimum Quantity	Integer	Minimum billable quantity	
Start Date	Date	Date the plan becomes effective	
End Date	Date	Date the plan expires	

Data Item	Type	Description	Comment
Auto-close	Boolean	Whether the plan closes automatically at term end	
Closable	Boolean	Whether the plan can be manually closed	
Pausable	Boolean	Whether the plan can be paused	
Renewable	Boolean	Whether the plan can be renewed	

Subscription Data Entity

Data Item	Type	Description	Comment
Subscription Number	Text	Unique identifier of the subscription	
Customer	Pointer	Customer entity	
Plan	Pointer	Recurring Plan entity	
Start Date	Date	Subscription start date	
Expiration Date	Date	Subscription end date	
Payment Terms	Text	Agreed payment terms	
Order Lines	Pointer	Product, Quantity, Unit Price, Taxes, Amount	May be several
Status	Category	Draft / Quotation / Confirmed / Active /	

Data Item	Type	Description	Comment
		Closed	

Quotation Template Data Entity

Data Item	Type	Description	Comment
Template Name	Text	Name of the template	
Validity Days	Integer	Number of days the quotation remains valid	
Recurring Plan	Pointer	Default Recurring Plan for this template	
Product Lines	Pointer	Default product lines pre-filled from this template	May be several

Invoice Data Entity

Data Item	Type	Description	Comment
Invoice Number	Text	Unique identifier of the invoice	
Customer	Pointer	Customer entity	
Subscription	Pointer	Source subscription	
Product Lines	Pointer	Line items with taxes and amounts	May be several
Total	Currency	Total invoice amount, including tax	

Data Item	Type	Description	Comment
Status	Category	Draft / Confirmed / Paid	

Payment Data Entity

Data Item	Type	Description	Comment
Invoice	Pointer	Invoice being settled	
Payment Method	Category	Method used to pay	
Amount	Currency	Amount paid	
Payment Date	Date	Date payment was received	

Discount Data Entity

Data Item	Type	Description	Comment
Discount Name	Text	Name of the discount rule	
Type	Category	Fixed / Percentage	
Minimum Purchase	Currency	Minimum order value required	
Minimum Quantity	Integer	Minimum quantity required	
Start Date	Date	Date the discount becomes valid	

Data Item	Type	Description	Comment
End Date	Date	Date the discount expires	
Limit Usage	Integer	Maximum number of times the discount may be used	
Applies To	Category	Products / Subscriptions	May be several

Tax Data Entity

Data Item	Type	Description	Comment
Tax Name	Text	Name of the tax rule	
Percentage	Decimal	Tax rate applied	
Type	Category	Classification of the tax	

3.3.2 Security

The server on which the Subscription Management System resides enforces role-based permissions so that only the Admin can access configuration screens for products, recurring plans, taxes, and discounts, and only the Admin can create Internal User accounts. Internal Users have operational read/write access to subscriptions, quotations, invoices, and payments, but no access to system configuration. Portal/Users have read-only access restricted to their own subscription, invoice, and payment records. Authentication is performed using email and password, with passwords required to exceed 8 characters and to contain uppercase, lowercase, and special characters. Password reset is

performed via a verified link sent to the registered email address. All sessions are authenticated, and sensitive fields such as password hashes are never displayed or transmitted in plain text.